

PROJECT		JOB NO.				
CONTRACTOR :						
SUB-CONTRACTOR : Winner Engineering Pte Ltd						
TRADE		: ACMV SERVICES	DRAWING NO :			
FUNCTIONAL - TESTING COMMISSIONING CHECKLIST (WATER COOLED CHILLED)						
Unit Designation :						
Unit Make :						
Model no.						
Location :						
Serial no :						
S/N	Section	Inspection / Test Item	CHECKED CONDITION (Please tick inside checkbox)			Remarks
			Yes	No	NA	
1	General	Check that the equipment is clean, free of debris, and undamaged before commencing operations.				
2	Documents	Verify chiller tag, make, model, serial number, refrigerant and rated capacity.				
3	Electrical	Verify supply voltage, phase sequence and running current on applicable phases.				
4	Controls	Test local start/stop, Auto/Manual, Remote/BMS and run/trip indications where provided.				
5	BMS	Compare BMS commands, statuses and analog readings with actual field conditions.				
6	OEM startup	Complete manufacturer-supervised startup and record startup report.				
7	Sequence	Verify CHW pump, CW pump, tower and isolation-valve startup sequence.				
8	CHW flow	Measure evaporator water flow and entering/leaving water pressure.				
9	CW flow	Measure condenser water flow and entering/leaving water pressure.				
10	Flow interlocks	Test evaporator and condenser low/no-flow permissives using manufacturer-approved simulation.				
11	Temperature	Record evaporator entering/leaving CHW temperatures and leaving-water setpoint.				
12	Temperature	Record condenser entering/leaving CW temperatures.				
13	Compressor	Record refrigerant pressures/temperatures, compressor load and oil readings where applicable.				
14	Cooling duty	Verify cooling capacity and input power at agreed test points.				
15	Efficiency	Verify kW/RT or COP where specified for acceptance.				
16	Loading	Test load/unload response, capacity control and current limiting where provided.				
17	Safety controls	Test applicable pressure, temperature, freeze, oil and electrical protections by approved simulation.				
18	Safety controls	Test emergency stop and refrigerant alarm/ventilation interfaces where specified.				
19	Plant control	Verify remote setpoint, demand limit and plant staging/duty-standby logic where provided.				

20	Shutdown	Verify normal stop, pump run-on, anti-recycle delay and restart permissives.				
21	Power recovery	Verify power-loss/recovery response under approved test procedure.				
22	Stability	Run at agreed duties for specified duration; check noise, vibration, leaks and external condensation.				
23	Completion	Record final setpoints, control mode and operating settings; restore normal operation.				

Comments to follow up Action :

Yes = satisfactory; No = rectify and re-inspect; NA = not applicable (state reason). Blank = not checked.

CHECKED BY:		WITNESSED BY	INSPECTED BY
NAME			
DESIGNATION :			
COMPANY :			
SIGNATURE :			
DATE :			